

Wildlife Corridors: Removing Barriers for Developers

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Developers and city planners looking to expand urban infrastructure often face an important problem: wildlife. Many developments don't make it past the proposal stage due to sensitive ecosystems, environmental regulations or public perception of loss of green space. Developments that take these restrictions into account in the planning stages are rewarded with cost-efficient, wildlife friendly solutions. Wildlife corridors exhibit the best of these qualities.

Keep reading to learn why corridors are important to ecosystem health, and excellent options for developers looking to improve public opinion.

By the numbers¹:

- Canada is home to over 70,000 wild species
- 1 in 5 households participated in unpaid conservation activities in 2013
- Legislation protecting endangered species can halt development at any stage
- The government levied over a million dollars in penalties to businesses for wildlife infractions
- Canadian businesses spent \$8.6 billion on environmental protection in 2006
- In 2006, 34% of business used at least one environmental management practice

¹www150.statcan.gc.ca

What are Wildlife Corridors?

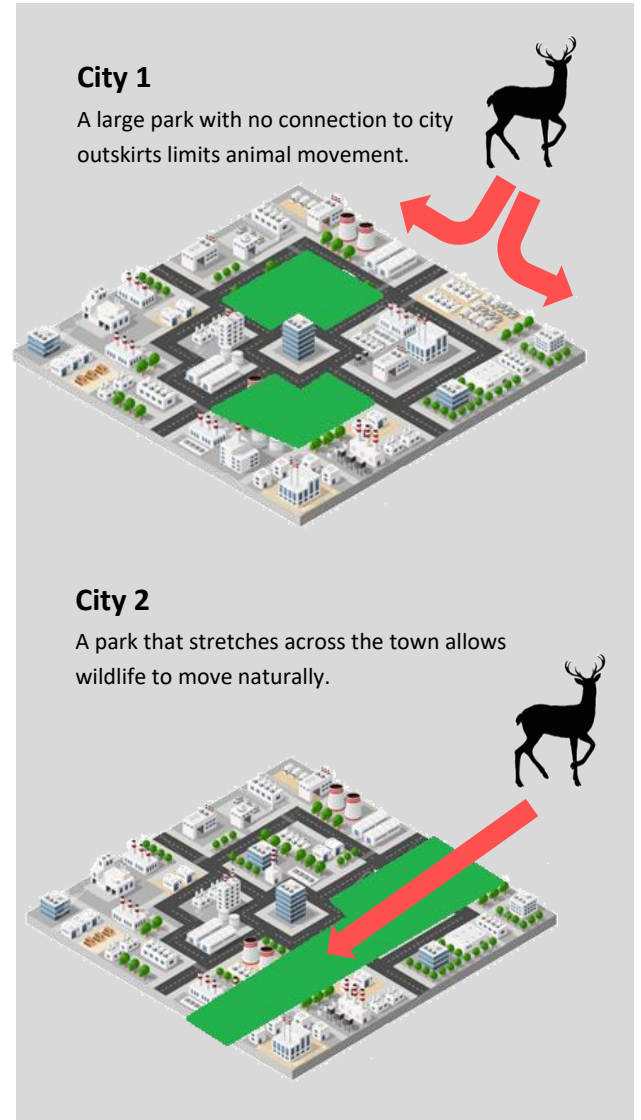
When habitats are interrupted by human development, it creates barriers to natural wildlife movement. Wildlife corridors create a link between natural areas, so that animals can migrate or move freely without entering urban environments.

Consider an example with two cities: City 1 has natural areas to the north and south, and a nice big park right in the middle. For animals to follow their natural migration patterns, they are forced to move into urban areas, becoming nuisances to humans, or to take efforts to find a way around.

City 2 has a park stretching all the way through the middle of the town, so that animals can make their way from one side of the city to the other without leaving their natural environment. This city has a wildlife corridor.

Corridors promote biodiversity for all the species in that area, which is a key factor in ecosystem health². When wildlife corridors are in place:

- animals are free to move and occupy new areas when resources are scarce, giving resources a chance to naturally replenish
- migrating species can move naturally without becoming nuisances to the human population
- animals can find new mates, increasing genetic diversity
- Wildlife does not become a nuisance to humans



How Developers Benefit from Wildlife Corridors

Including a wildlife corridor in your development not only improves ecosystem health, it eliminates objections from government agencies and public stakeholders who may be concerned about wildlife impacts.

² https://en.wikipedia.org/wiki/Wildlife_corridor

Types of Wildlife Corridors

Wildlife corridors come in all shapes, sizes and budgets. Whether your development is a suburb, an industrial project or a single building, there is a corridor for your needs! Corridors are broken down into four categories, by size: Regional, Sub-Regional, Local and Sub-Local, described in the table below³.

	Scale	Purpose	Example
Regional	> 500 m wide	Acts as a habitat itself, but also connects large natural areas	The Greenbelt in the Greater Toronto Area
Sub-Regional	50 – 300 m	Wildlife doesn't live in these corridors, but they are used for movement	Banff National Park Wildlife Passes
Local	10 - 50 m	Smaller, less defined corridors which provide a connection to natural areas	A creek running through a town
Sub-Local	10 m or less	Very small spaces that provide opportunities for wildlife movement	A line of hedges surrounding a property

Within these categories, there is a myriad of corridor types. Features which allow movement of wildlife from any size and species is considered a wildlife corridor! Examples include:

- Bridges
- Tunnels
- Creeks and waterbodies
- City parks
- Shrubs or spaces with natural vegetation
- Rooftop gardens

The Downside

Measuring corridor success is difficult. The best way to measure success is with ongoing monitoring and assessment. This can range in cost and duration, depending on the project.

Wildlife may not always be receptive to a certain type of corridor. For example, animals which are frightened by highway noise may be reluctant to cross by tunnel but favour a bridge with no enclosed sides and a clear view of surroundings. To avoid these pitfalls, it is highly recommended that an experienced environmental engineering company is involved in the planning, construction and monitoring of all corridors.



In Photos:

1. Crabs migrate over a highway in Christmas Island National Park (Australia)
2. An endangered butterfly migrates through St. Francis' Satyr garden (USA)
3. A deer crosses Highway 43 through a tunnel (Montana, USA)
4. Elephants travel in Terai Arc Landscape, a series of linked national parks (Nepal/India)

³ <http://www.environment.nsw.gov.au/resources/nature/landholderNotes15WildlifeCorridors.pdf>

Case Study: Banff National Park⁴

Banff National Park faces problems balancing the needs of development for a growing tourism industry with federal requirements for conserving wildlife in a pristine national park.

A major highway cuts through the center of the pass. With increased traffic on that highway, there were wildlife fatalities and large carnivores were not able to follow natural patterns of movement, creating an imbalance in the ecosystem, and concerns from federal policy makers.

To mitigate these problems, 40 animal passes were constructed. The passes consist of two types: bridges over the highway and underpass tunnels. Both have native vegetation and are constructed to look like natural features.

Motion cameras on the passes have revealed the project to be successful. Highway mortality is down 80% and 11 species of large mammal regularly use the passes.



Not only has the project been a success for wildlife conservation, Banff National Park is now internationally recognized for its innovation, with visitors coming from around the world specifically to see the now-famous wildlife corridors.

For developers, this is an ideal scenario: the highway no longer faces threats from legislators, drives more traffic than ever before, and is heralded as an environmental success story!

⁴ <https://conservationcorridor.org/2012/10/banff-national-park/>

Conclusion

In conclusion, wildlife corridors are an excellent solution for developers. Many developers face barriers from conservationists, concerned citizens or government legislation that prevents development due to wildlife constraints. By creating wildlife corridors, developers can mitigate those concerns and improve public perception of their projects. Government legislation is increasingly moving towards conserving wildlife, and taking steps to move with this trend will benefit your organization.

Examples like the Banff National Park wildlife passes are proof that wildlife corridors not only work for conservation, but they also encourage development projects and can be a point of celebration for development with nature, rather than outrage over urbanization.

There are many examples of large and small scale corridors. There is an appropriate corridor option no matter how big or small your project is.

Visit www.fakeenvironmentalconsultingcompany.com to learn more about which corridor options may be right for you.



Fake Environmental Consulting Inc. has been engineering wildlife corridors and green spaces for over a decade.

Specializing in sub-regional and local corridors, Fake Environmental has an extensive portfolio and a team of experts who would love to discuss your unique project.